

University American College Skopje
School of Computer Science and Information Technology

Academic 2024/25

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Course Code: Databases 2 (CSCI 3511)
Semester: IV semester, 2026/2027
Year: Fall, 2025
Prerequisites: CSCI 2331, CSCI 2511
Lecturer: Venera Gudachi, PhD

Assistant: Zoran Gichevsski BsC
Duration of Course: 45 hours
Credit Hours: 6 ECTS
Schedule hours: Monday, 15:30- 16:30 – lectures
Monday, 16:30-18:30 – exercises
Room: L2 / 2nd floor
Consultations hours: Every last Monday of the month from 16:30 hrs.

Instructor's contact details:

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Required text:

Principles of Distributed Database Systems, 4th edition

M. Tamer Ozsu, Patrick Valduriez

Paperback: 681 pages

Publisher: Springer Nature Switzerland AG 2020 4th edition

Language: English

ISBN 978-3-030-26252-5

ISBN 978-3-030-26253-2 (eBook)

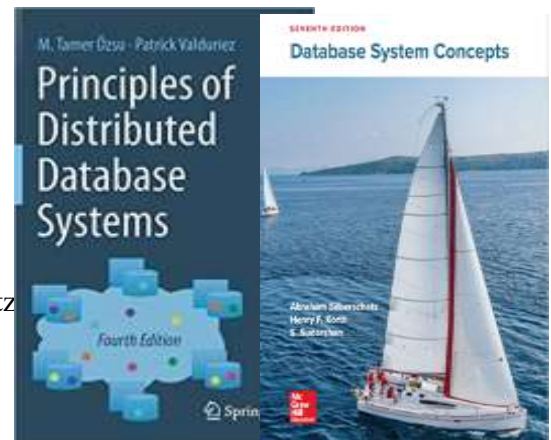
Database System Concepts, 7th Edition, by Abraham Silberschatz

Henry F. Korth S. Sudarshan

McGraw Hill Education, 2020

ISBN: 978-0-07-802215-9

Books are available in Moodle (Course book)



Course outline/ description:

This course is designed as an extension of Databases 1, bridging the gap between traditional database management systems (DBMS) and the emerging complexities of distributed databases. The course will explore advanced topics in database performance optimization, transaction management, query processing, and security. Furthermore, students will be introduced to the basics of distributed databases, including data fragmentation, replication strategies, and distributed transactions.

Students will learn how to enhance relational database systems and prepare for the challenges of distributed systems. Practical knowledge will be developed through SQL query optimization, advanced transaction handling, and the design of distributed data architectures. By the end of the course, students will have a strong foundation for both managing traditional databases and understanding the intricacies of distributed databases.

Learning objectives:

- Understand advanced concepts in relational database design, including normalization, entity-relationship modeling, and relational algebra.
- Develop proficiency in SQL for managing and querying relational databases, including advanced queries, subqueries, and joins.

- Implement database optimization techniques such as indexing, query optimization, and execution plan analysis.
- Design and manage database transactions, ensuring data integrity and consistency through concurrency control.
- Utilize constraints, triggers, and stored procedures to enhance database functionality and enforce data rules.
- Apply normalization techniques to reduce redundancy and improve database performance, achieving higher normal forms (3NF, BCNF, etc.).
- Analyze and repair database structures using dependency inheritance and reparation techniques.
- Develop secure database systems by implementing authentication, authorization, and encryption techniques.
- Create, maintain, and manage database views and indexes to optimize data access and presentation.
- Build and implement real-world database applications, integrating the theoretical knowledge into practical use cases.
- Evaluate different database strategies to improve performance, scalability, and security in various real-world scenarios.

Course delivery and format

The course will be conducted on-site, with exceptions permitted upon approval. The course will combine lectures, guided exercises, laboratory practice, and AI-assisted learning activities to support both theoretical understanding and practical skills. It is designed to help students develop expertise in advanced SQL programming, database performance optimization, and modern database architectures.

Lectures. Core concepts will be introduced through instructor-led presentations supported by slides and real-world examples. Topics will include advanced SQL constructs (e.g., triggers, stored procedures, functions, and views), query optimization strategies, transaction management and recovery, indexing methods, concurrency control, distributed databases, and security models. Lecture outlines will be shared to guide students in preparing for discussions and exercises.

Practical exercises. Students will solve progressively complex SQL problems, perform query tuning, design indexing strategies, and implement transactions with concurrency control mechanisms. Exercises will also include normalization of large datasets and design of relational schemas. Students will apply theory using MS SQL Express with SQL Management Studio, MySQL with MySQL Workbench, and MongoDB with Mongoshell. Projects will focus on implementing advanced SQL features, analyzing query execution plans, testing recovery scenarios, and developing database-driven applications. Instructor guidance will be available throughout.

Interactive Learning with Mentimeter. To encourage engagement and immediate feedback, Mentimeter will be used during lectures for live polls, knowledge checks, and short case-based discussions. This tool will help students reflect on complex database topics in real time and foster active participation.

AI-Assisted Teaching. Students will be encouraged to use Large Language Models (LLMs) (e.g., ChatGPT, Gemini, Claude or similar tools) to generate SQL query examples, experiment with optimization strategies, or clarify advanced theoretical concepts. AI-generated study aids (summaries, code snippets, or mini-quizzes) may also be provided to strengthen student engagement with complex topics.

AI-Assisted Surveys and Assignments. Optional AI-assisted self-tests will be offered throughout the semester to allow students to evaluate their progress on advanced database topics such as indexing, query plans, and recovery protocols. These will not affect the final grade but will support formative assessment.

AI-Powered Support. An AI-driven Q&A platform may be made available for instant explanations and query debugging tips. Students will be encouraged to critically assess AI outputs, particularly in complex SQL optimization scenarios, to build professional judgment and awareness of limitations.

Official Quizzes in Moodle. Periodic weekly online quizzes will assess mastery of advanced SQL and database management topics. These will be automatically graded and contribute to the active participation grade.

Independent Study. Students are expected to review academic literature, technical documentation, and advanced database references. Independent work will include reading research papers on database systems, experimenting with SQL performance tuning, and preparing case studies for class discussion.

Learning outcomes

By the end of this module, students will achieve the following learning outcomes:

- **Knowledge and understanding:** Students will develop a comprehensive understanding of advanced database systems, including key concepts such as relational models, SQL, data normalization, query optimization, indexing, and transaction management. They will gain insight into the principles of database design, including constraints, triggers, views, functional dependencies, and security measures. Students will also understand the potential role of AI/LLM-based tools in supporting database learning and practice, including their benefits and limitations

- **Skills:** Students will acquire practical skills in designing and managing relational databases. They will learn to create efficient database structures, write complex SQL queries for data retrieval and manipulation, and optimize database performance. Through hands-on exercises and projects, they will apply these skills to solve real-world database challenges, including securing and maintaining data integrity. Students will also develop skills in responsibly applying AI/LLM tools for drafting, testing, and evaluating database solutions.
- **Competences:** Students will be able to critically evaluate different database design strategies, applying theoretical concepts to improve database performance, scalability, and security. They will demonstrate competence in implementing and managing database systems, and develop effective communication and collaboration skills by presenting project findings, writing reports, and engaging in peer reviews. Through project-based learning, students will also foster their ability to engage in lifelong learning and prepare for future professional challenges in the field of database management. Competence will also include the ability to use AI tools as supportive instruments in their learning and professional practice, while maintaining independent critical judgment and adherence to ethical standards.

Course delivery plan:

Class	Month	Lecture Topics	Exercises Topics
1	Week 1	Introduction to Advanced Database Systems	Introduction
2	Week 2	Advanced Database Design and Data Modeling	Exercises on the subject
3.	Week 3	Database Integration and Multidatabase Systems	Exercises on the subject
4.	Week 4	Parallel Database Systems	Exercises on the subject
5.	Week 5	Peer-to-Peer Data Management	Exercises on the subject
6.	Week 6	Web Data Management	Exercises on the subject
7.	Week 7	Mid Term Exam	Mid Term Exam
8.	Week 8	Big Data: Concepts, Architectures and Processing	Exercises on the subject
9.	Week 9	NoSQL Databases	Exercises on the subject
10	Week 10	NewSQL Databases	Exercises on the subject
11.	Week 11	Polystores and Multi-Model Databases	Exercises on the subject
12.	Week 12	Modern Data Storage and Processing Architectures	Exercises on the subject
13.	Week 13	Database Project Development	Exercises on the subject
14.	Week 14	Project presentation / Trial	Exercises on the subject
15.	Week 15	Reading week	Reading week
16	Week 16	Final exam	Final Exam

Instructor's expectations from students/Students' responsibilities:

- Regular attendance of lectures and exercises on-site.
- Timely arrival at classes.
- Careful follow-up of lectures and exercises.
- Academic behavior and preservation of the working atmosphere of classes.
- Mandatory remarks about everything that is written and spoken in classes.
- Active participation in the discussion of the lessons by completing the provided quizzes.
- Timely preparation and mandatory oral presentation of semestral projects if given.
- Independence and honesty in work.
- Avoid anti-social behavior in college and anti-academic behavior in the classroom (i.e., plagiarism, cheating, etc.)
- Impose a self – discipline regarding college's rules and procedures.
- If online sessions are scheduled on special occasions, the students are required to turn On their cameras at all times during the class.

Evaluation structure and assessments during the teaching semester:

Grading Criteria		
Grading	% Qty	Points
A-	96-100	4.00
A	90-95	3.67
B	83-86	3.00
B-	80-82	2.67
C+	77-79	2.33
C	73-76	2.00
C-	70-72	1.67
D+	67-69	1.33
D	63-66	1.00
D-	60-62	0.67
F	0-59	0.00

Categories in the assessment:

1. Attendance - 10%
2. Activity/Participation - 10%
3. Assignment/ project- 20% (assignment 10 and project 10)
4. Midterm Exam – 30 % and
5. Final Exam 30%

Examinations will take place on-site without exceptions. All work should be performed in lab computers (no personal laptops are allowed). Examinations in labs will be performed in Safe Exam Browser.

1. **Attendance (10%):** Up to 25% absences will still allow students to earn points for attendance. For absences between 25-50%, no attendance points will be awarded, but students will still be permitted to take the Final Exam. If a student exceeds 50% absences, they will not be allowed to take the Final Exam. Absences must be approved by both the lecturer and the administration, with mutual notification.
2. **Activity/Participation (10%):** Engaging in class discussions, group activities, and practical exercises will positively impact the participation grade. Contributions through interactive tools like Mentimeter will also be considered part of a student's participation. Continuous and meaningful involvement in class activities will be rewarded as part of the grading criteria.
3. **Assignment/ project (20%):** The assignments will bring 10% and project 10%, both will be focusing on applying the concepts learned throughout the course.
4. **Midterm Exam (30%):** Will be covering both theoretical concepts (multiple-choice, essay, and true/false questions) and practical problems related to the course material.
5. **Final Exam (30%):** Will be including questions on theory (multiple-choice, essay, and true/false) as well as practical problems requiring the application of the tools and techniques taught in the course.

IMPORTANT:

- The summative grades and minimum required performances are a minimum 20% of the Midterm Exam and Final Exam points.
- The distribution between theoretical and practical part is 40% theory and 60% practice. Meaning that to be able to pass the exam in each category in the assessment there is a need to obtain at least 20% both in theoretical and practical part.

Evaluation structure and assessments for Make-Up examinations

Examinations will take place on-site without exceptions. All work should be performed in lab computers (no personal laptops are allowed). Examinations in labs will be performed in Safe Exam Browser.

Categories in the assessment:

- Theoretical part -40%
- Practical part -40%
- Assignment/ project- 20%

No other categories will be taken into consideration such as attendance and activity/participation

3) Part-time students

The exam will include questions on the theory taught (multiple-choice, T/F and Critical response Essay). Whereas in the practical part will include exercises from the material.

Categories in the assessment:

- Theoretical part 50%
- Practical part 50%

Exams are closed books. The professor has the right to contact the student to take an extra-oral exam if needed to determine the final grade.

Mobile phones are strictly not tolerated in the class for any use (including computations). Active participation is meant as the effort and the interest that a student shows in the class, including homework.

After each session students are expected to study all the relevant materials, read all the associated exercises, identify the difficult points and pose their questions in the next session either directly to me or in the class. Cheating and plagiarism in any form will result immediately in the grade F. The grade of (incomplete) is assigned

only when a student misses the Midterm or the Final exam due to a **college-excused absence**, and is unable to make up before final grades, and are due to the Record's Office. Please, consider seriously that there will be no exceptions concerning the above policy.

Class Conduct

Students are held responsible for everything that is announced, presented or discussed in class. The way to avoid any misunderstanding associated with this course is to attend classes. Students are expected to attend class regularly since attendance records will be kept. Late arrivals create distraction resulting in decline in the quality of the lectures.

Importantly, coming late to class is also disruptive to other student's colleagues and will be tolerated only up to five minutes late at the beginning of the class. Please, refrain from talking during class, it is disruptive to everyone. The best behavior from all students is expected as that what education is all about. In case there is a question about the material, please don't hesitate to ask the instructor and/or the assistant instructor; answering questions is part of our job. In case someone feels lost or confused, making questions may help clarify the topic. Please, consider that the language of instruction is English, so all our conversation into the class must be in this language.

Keep in mind the following

C- or better is required to use a course either as a prerequisite or as a major requirement.

G.P.A. (Grade Point Average): is computed for each student using the quality points earned for each course taken.

A G.P.A of at least 1.67 is required for transfer and to graduate from ACS.

Academic Honesty

Academic honesty is a core value that safeguards the reputation of University American College Skopje and its faculty, staff, and students.

All UACS members are expected to conduct themselves in an honest manner observing the University Code of Ethics. Plagiarism, cheating, multiple submissions, falsification of data, and unauthorized assistance are considered as serious offences and may result in grade F (fail) on the exam, project, and course, and/or lead to further sanctions, including expulsion from the University. Any work that is submitted at any stage must be your own. Any words, ideas, or data borrowed from other sources must be properly credited and documented.

You may not use generative AI tools, such as ChatGPT, while completing any assignment, unless explicitly permitted by the course instructor in specific contexts. Any use of generative AI tools outside of these parameters constitutes plagiarism and will be sanctioned accordingly.

Late Work

Many reasons are given for late work (e.g., flat tires, printer problems, illness, oversleeping, etc.). No matter what the reason, the bottom line is that either (a) the work was completed and turned in when required or (b) the assignment was not completed and turned in when required.

As specified earlier, assignments are due in class at the beginning of class on the specified due date. There is NO PROVISION for late work on any assignment (i.e., late work is not accepted). It is strongly recommended to have backup systems in place so that students can have all work completed on schedule. Having the work completed on schedule is a key to early success in student's business career.

NOTE: Late submissions are not accepted. Partial credit will NOT be given for late work.

Wishing you an interesting and creative academic semester.

Prepared by: Venera Gudachi, Ph.D. and Zoran Gichevski, BsC